QM4 Lecture Notes

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Abstract

This is my lecture note on QM4 in EPFL for 2026 spring semester.

Contents

1 Lecture 5: Gauge Invariance of QM

1.1 Classical Electromagnetism

1.1.1 Potential Convention

We here set up our convention for classical EM. The two Maxwell's equations that are source independent can be satisfied if we use a potential ansatz:

$$B(x,t) = \nabla \times A(x,t), \quad E(x,t) = -\nabla\varphi(x,t) - \partial_t A(x,t) \tag{1}$$

Note that due to the fact QM is not covariant theory, we will not use the 4-vector notation which might make life complicate. One can show that the definition is non-unique yet can be related by a gauge transformation:

$$\varphi_\chi = \varphi - \partial_t \chi, \quad A_\chi = A + \nabla\chi \tag{2}$$

1.1.2 Electrodaynamics Lagrangian

Then if we have a charged particle in the EM field, the Lagrangian is given by:

$$L(x,\dot{x},t) = \frac{1}{2}m\dot{x}^2 - e\varphi(x,t) + e\dot{x}\cdot A(x,t) \tag{3}$$

one can prove that the virational principle gives us the EoM with lorentz force. We can see that the Lagrangian is not gauge invariant:

$$L_\chi = L + e(\partial_t\chi - \dot{x}\cdot\nabla\chi) = L + e\frac{d}{dt}\chi(x(t),t) \tag{4}$$

> **Remark**
>
> Note that here is a total derivative of χ, not a partial derivative!!

Yet the action is gauge invariant up to a boundary term:

$$S_\chi = S + e\big(\chi(x_f,t_f) - \chi(x_i,t_i)\big) \tag{5}$$

1.1.3 Electrodynamics Hamiltonian

Then in order to do canonical quantization, we need to find the Hamiltonian. The canonical momentum is given by:

$$p = \frac{\partial L}{\partial \dot{x}} = m\dot{x} + eA(x,t) \tag{6}$$

this is different from the commonly defined "QM" momentum $\pi = m\dot{x}$. The Hamiltonian is then given by:

$$H = p\dot{x} - L = \frac{1}{2m}(p - eA(x,t))^2 + e\varphi(x,t) \tag{7}$$

We can see that the Hamiltonian is also not gauge invariant:

$$p_\chi = p + e\nabla\chi, \quad H_\chi = H - e\partial_t\chi \tag{8}$$

1.2 QM with EM Field

1.2.1 Quantization

Then we can try to define a canonical quantized theory of this. Note that the quantum theory is not unique, for the ordering of the operators shall be defined. We define the following Hamiltonian:

Definition 1.1 **QED Hamiltonian**

We use the following Hamiltonian for quantization:

$$\hat{H} = \frac{1}{2m}(\hat{p}^2 - e(\hat{p} \cdot A(x,t) + A(x,t) \cdot \hat{p}) + e^2 A(x,t)^2) + e\varphi(x,t) \tag{9}$$

note that $A(\hat{x}, t)$ is a time dependent function of the space operator now and $\hat{p}$ which is the canonical momentum operator.

We then impose the canonical commutation relation between operators:

Definition 1.2 **Canonical Commutation Relation**

We impose the following canonical commutation relation:

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \quad [\hat{x}_i, \hat{x}_j] = 0, \quad [\hat{p}_i, \hat{p}_j] = 0 \tag{10}$$

note that with this commutation relation, it is p that is the canonical momentum operator and $p = -i\hbar\nabla$ in the position basis. We can also define a Mechanical Momentum operator:

$$\hat{\pi} = \hat{p} - eA(\hat{x}, t) \tag{11}$$

This doesn't satisfy the canonical commutation relation, but it is a physical observable as we will see later.

Then we can also do path integral quantization. The transition amplitude is given by:

$$G(x_f, t_f; x_i, t_i) = \int_{x_i}^{x_f} \mathcal{D}[x] \exp\left(\frac{i}{\hbar} S[x(t)]\right) \tag{12}$$

1.2.2 Gauge Transformation of Operators

We can directly see that the Hamiltonian and the Action are not gauge invariant. Yet we want the theory to be gauge invariant. One requirement is that **the schrodinger equation should be gauge invariant**. This means that the wavefunction should also transform under gauge transformation:

$$i\hbar\partial_t \left|\psi_\chi(t)\right\rangle = \hat{H}_\chi(t) \left|\psi_\chi(t)\right\rangle \tag{13}$$

In order to have this equation to be well-defined, we must first define what O_χ means. Naively, we can just substitute the classical gauge transformation of the observable into operators. Yet this may cause problems with the Momentum Operator. if we have:

$$p_\chi \sim ???? \sim p + e\nabla\chi \tag{14}$$

then the canonical momentum operator will not commute with each other after a gauge transformation, which is not consistent with the canonical quantization result. In order to preserve the canonical commutation relation, we take the following definition for gauge transformation of momentum operator:

$$\hat{p}_\chi = \hat{p}, \quad \hat{\pi}_\chi = \hat{\pi} - e\nabla\chi(\hat{x}, t) \tag{15}$$

One should note that with this definition, the lagrangian and Hamiltonian might change differently from the classical case in the operator form.

1.2.3 Gauge Transformation of Wave Function

In order to preserve this form of schrodinger equation:

$$i\hbar\partial_t \left|\psi_\chi(t)\right\rangle = \hat{H}_\chi(t) \left|\psi_\chi(t)\right\rangle \tag{16}$$

We can have the following transformation rule of the wavefunction:

$$\left|\psi_\chi(t)\right\rangle = U_\chi(t) \left|\psi(t)\right\rangle \quad U_\chi(t) = \exp\left(\frac{ie}{\hbar}\chi(\hat{x}, t)\right) \tag{17}$$

If we write this gauge transformation in the position basis, we have:

$$\psi_\chi(x, t) = e^{\frac{ie}{\hbar}\chi(x,t)}\psi(x, t) \tag{18}$$

1.2.4 Gauge Transformation of Propagator

Then we can also see how the propagator transforms under gauge transformation. We know that the action transforms as Equation 5, so the propagator transforms as:

$$G_\chi\left(x_f, t_f; x_i, t_i\right) = e^{\frac{ie}{\hbar}\left(\chi\left(x_f,t_f\right)-\chi\left(x_i,t_i\right)\right)} G\left(x_f, t_f; x_i, t_i\right) \tag{19}$$

This is also consistent with the fact that the wave function can be written in terms of the propagator:

$$\begin{aligned}
\psi\left(x_f, t_f\right) &= \int d^3x_i G\left(x_f, t_f; x_i, t_i\right)\psi\left(x_i, t_i\right) \\
\psi_\chi\left(x_f, t_f\right) &= \int d^3x_i G_\chi\left(x_f, t_f; x_i, t_i\right)\psi_{\chi\left(x_i, t_i\right)} \\
&= \int d^3x_i e^{\frac{ie}{\hbar}\left(\chi\left(x_f,t_f\right)-\chi\left(x_i,t_i\right)\right)} G\left(x_f, t_f; x_i, t_i\right) e^{\frac{ie}{\hbar}\chi\left(x_i,t_i\right)}\psi\left(x_i, t_i\right) \\
&= e^{\frac{ie}{\hbar}\chi\left(x_f,t_f\right)}\psi\left(x_f, t_f\right)
\end{aligned} \tag{20}$$

1.2.5 U-Picture

We before make the wave function transform under gauge transformation. As Schrodinger picture to Heisenberg picture, in fact we can also define a "U-picture" where the operators transform under gauge transformation with an additional $U_\chi(t)$ operatro and the wave function is invariant.

We can define the following transformation of the operator:

$$O_\chi^{(U)}(t) = U_\chi^\dagger(t) O_\chi(t) U_\chi(t) \tag{21}$$

where $O_\chi(t)$ is naively subsituting the classical gauge transformation into the operator.

Note that consider the expectation value of an operator, if we have:

$$\hat{O}_\chi^{(U)}(t) = \hat{O}(t) \tag{22}$$

Then we view the observable as gauge invariant. We take all these observables to be the physical observables of the theory. In this way, we can have a gauge invariant quantum theory.

There are some examples of this transformation:

$$
\begin{aligned}
&\hat{x}_\chi = \hat{x} = \hat{x}_\chi^{(U)}, \\
&\hat{p}_\chi = \hat{p}, \qquad\qquad \hat{p}_\chi^{(U)} = \hat{p} + e\nabla\chi(\hat{x}, t), \\
&\hat{\pi}_\chi = \hat{\pi} - e\nabla\chi(\hat{x}, t), \qquad\qquad \hat{\pi}_\chi^{(U)} = \hat{\pi}
\end{aligned}
\tag{23}
$$

Thus we can also see here that the mechanical momentum is physical observable while the canonical momentum is not, though the canonical momentum form is gauge independent.

1.2.6 Probability Density and Current

In basic QM, we have the probability density and current defined as:

$$
\rho(x,t) = |\psi(x,t)|^2, \quad J(x,t) = \left(\frac{1}{2m}\right)(\psi^*(-i\hbar\nabla)\psi + c.c.) \tag{24}
$$

They satisfy the conservation law:

$$
\partial_t \rho + \nabla \cdot J = 0 \tag{25}
$$

However, in the presence of EM field the probability current in this form will be not gauge invariant. We can define a gauge invariant probability current.

> **Definition 1.3** **Gauge Invariant Probability Current**
>
> We can define the following gauge invariant probability current:
>
> $$
> J(x,t) = \left(\frac{1}{2m}\right)(\psi^*(-i\hbar\nabla - eA(x,t))\psi + c.c.) \tag{26}
> $$

Then with the probability density defined as $\rho(x,t) = |\psi(x,t)|^2$, we can also show that the conservation law is still satisfied:

$$
\partial_t \rho + \nabla \cdot J = 0 \tag{27}
$$

2 Lecture 6: Landau Levels

We now consider the eigenstates and eigenvalues of Hamiltonian of a charged particle in a magnetic field. We will see that the energy levels are quantized, and each level has a large degeneracy. These quantized energy levels are called Landau levels.

2.1 Magnetic Translation Operator

2.1.1 Translation Operator

In basic QM, we often define a **Unitary Operator** as the translation operator satisfying:

$$T_{r_0}|r\rangle = |r + r_0\rangle \tag{28}$$

A realization of this operator is given by:

$$T_{r_0} = \exp\left(-\frac{i}{\hbar} r_0 \cdot \hat{p}\right) \tag{29}$$

An operator is called **Translation Invariant** if it commutes with the translation operator:

$$\langle r|O|r\rangle = \langle r + r_0 \; |O|r + r_0\rangle \quad \Leftrightarrow \quad T_{r_0}^{\dagger} O T_{r_0} = O \tag{30}$$

Notice that the commutation relation between the translation operator and the position and momentum operator is given by:

$$T_{r_0}^{\dagger} \hat{x} T_{r_0} = \hat{x} + r_0, \quad T_{r_0}^{\dagger} \hat{p} T_{r_0} = \hat{p} \tag{31}$$

One can also define a translation operator in the momentum space:

$$Q_{p_0} = \exp\left(\frac{i}{\hbar} p_0 \cdot \hat{x}\right), \quad Q_{p_0}|p\rangle = |p + p_0\rangle \tag{32}$$

Its commutation relation with the position and momentum operator is given by:

$$Q_{p_0}^{\dagger} \hat{x} Q_{p_0} = \hat{x}, \quad Q_{p_0}^{\dagger} \hat{p} Q_{p_0} = \hat{p} + p_0 \tag{33}$$

In fact if we compare the definition with the gauge transformation operator. Then we see the momentum translation operator is just a gauge transformation operator with a linear gauge function:

$$Q_{p_0} = \exp\left(\frac{i}{\hbar} p_0 \cdot \hat{x}\right) = \exp\left(i \frac{e}{\hbar} \chi(\hat{x})\right), \quad \chi(x) = \frac{1}{e} p_0 \cdot \hat{x} \tag{34}$$

2.1.2 Magnetic Translation Operator

If we consider a electron gass in a magnetic field, using the **Symmetry Gauge**:

$$A(x) = \frac{1}{2} B \times x \tag{35}$$

then if the magenetic field is along the z direction, we have:

$$\hat{H} = \frac{1}{2m}\left[\left(\hat{p}_x + \frac{eBy}{2}\right)^2 + \left(\hat{p}_y - \frac{eBx}{2}\right)^2 + \hat{p}_z^2\right] \tag{36}$$

the translation operator is not a symmetry of the system anymore [doesn't commute with the Hamiltonian]. However, we can introduce a **Magnetic Translation Operator**:

Definition 2.1 **Magnetic Translation Operator**

For translation in the x direction, we have:

$$\hat{T}_{x;x_0} := \hat{T}_{(x_0,0,0)}\hat{Q}_{(0,eBx_0/2,0)} = e^{-\frac{i}{\hbar}x_0\hat{p}_x}e^{\frac{ieBx_0}{2\hbar}\hat{y}}, \tag{37}$$

and for translation in the y direction, we have:

$$\hat{T}_{y;y_0} := \hat{T}_{(0,y_0,0)}\hat{Q}_{(-eBy_0/2,0,0)} = e^{-\frac{i}{\hbar}y_0\hat{p}_y}e^{-\frac{ieBy_0}{2\hbar}\hat{x}} \tag{38}$$

Then one can check that the magnetic translation operator commutes with the Hamiltonian:

$$\hat{T}^\dagger_{x;x_0}\hat{H}\hat{T}_{x;x_0} = \hat{H}, \quad \hat{T}^\dagger_{y;y_0}\hat{H}\hat{T}_{y;y_0} = \hat{H} \tag{39}$$

Remark

One can understand this as performing a translation then perform a gauge transformation. Due to the fact that momentum translation operator is a gauge transformation.

2.1.3 Generators of Magnetic Translation

For these symmetry operators, we can also define the generators.

- For translation in the x direction, we have:

$$t_x = \hat{p}_x - \frac{eBy}{2} \quad \hat{T}_{x;x_0} = \exp\left(-\frac{i}{\hbar}x_0t_x\right) \tag{40}$$

- For translation in the y direction, we have:

$$t_y = \hat{p}_y + \frac{eBx}{2} \quad \hat{T}_{y;y_0} = \exp\left(-\frac{i}{\hbar}y_0t_y\right) \tag{41}$$

- Of course we also have the translation in the z direction:

$$t_z = \hat{p}_z \quad \hat{T}_{z;z_0} = \exp\left(-\frac{i}{\hbar}z_0t_z\right) \tag{42}$$

In a general form we have:

Definition 2.2 **Generator of Magnetic Translation**

$$t = \hat{p} + eA(\hat{x}) \quad \left[t, \hat{H}\right] = 0 \tag{43}$$

2.1.4 Path Integral of Magnetic Translation

In the symmetry gauge, the lagrangian can be written as:

$$L = \frac{1}{2}m\dot{x}^2 + \frac{eB}{2}(x\dot{y} - y\dot{x}) \tag{44}$$

We consider a transition amplitude from $|r_i\rangle$ to $|r_f\rangle$ in time t_i, t_f. Which is:

$$G(r_f, t_f; r_i, t_i) = \int_{r_i}^{r_f} \mathcal{D}[r]\exp\left(\frac{i}{\hbar}S[r(t)]\right) \tag{45}$$

If we perform a shift of the initial and final position:

$$G(r_f + r_0, t_f; r_i + r_0, t_i) = G(r_f, t_f; r_i, t_i)\exp\left(i\frac{eB}{\hbar}(x_0(y_f - y_i) - y_0(x_f - x_i))\right) \tag{46}$$

we compare with the form of the gauge transformation of propagator Equation 19, we can see that translating the initial and final position is equivalent to performing a gauge transformation with a linear gauge function:

$$\chi(x) = B(x_0 y - y_0 x) \tag{47}$$

we can see that this gauge transformation gives the operator that is exactly the extra operator in the definition of magnetic translation operator.

2.1.5 Non-Commutativity of Magnetic Translation

Now we want to translate a state around a small loop. We can do this by performing:

$$\hat{T}_{y;-y_0}\hat{T}_{x;-x_0}\hat{T}_{y;y_0}\hat{T}_{x;x_0}|r\rangle \tag{48}$$

Due to the extra gauge transformation, we have:

$$\hat{T}_{y;-y_0}\hat{T}_{x;-x_0}\hat{T}_{y;y_0}\hat{T}_{x;x_0}\ |r\rangle = \exp\left(-ieB\frac{x_0 y_0}{\hbar}\right)\ |r\rangle \tag{49}$$

We can define the quantum flux through the loop as:

$$\Phi_0 = \frac{2\pi\hbar}{e} \tag{50}$$

Then the state will acquire a phase factor:

$$\exp\left(-ieB\frac{x_0 y_0}{\hbar}\right) = \exp\left(-i2\pi\left(\frac{\Phi}{\Phi_0}\right)\right) \tag{51}$$

2.1.6 Magnetic Translation Generator as a Conserved Quantity

In Classical mechanics, the motion of a charged particle in a magnetic field has a conserved quantity:

$$t = m\dot{x} - ex \times B \tag{52}$$

We know that the mechanical momentum when doing quantiztion gives $\pi = p - eA(x)$. Then we can also define a quantum version of this conserved quantity:

$$t = \hat{p} - eA(\hat{x}) - e\hat{x} \times B \tag{53}$$

Notice that in symmetry gauge we have $A(x) = \frac{1}{2}B \times x$, then we have:

$$t = \hat{p} + eA(\hat{x}) \tag{54}$$

which is exactly the generator of magnetic translation Definition 2.2. Thus, everything is very consistent.

2.2 Landau Levels

2.2.1 Symmetry Gauge

Now we consider the eigenstates and eigenvalues of the Hamiltonian of a charged particle in a magnetic field. We first use the symmetry gauge to write the Hamiltonian:

$$\hat{H} = \frac{1}{2m}\left[\left(\hat{p}_x - e\hat{A}_x\right)^2 + \left(\hat{p}_y - e\hat{A}_y\right)^2\right] = \frac{\hat{\pi}_x^2 + \hat{\pi}_y^2}{2m} \tag{55}$$

Remark

Here we just neglect the translation in the z direction, which is just a free particle and does not contribute to the interesting physics of Landau levels.

To solve this Hamiltonian, a trick is to perform a canonical transformation. We define the following operators:

$$\hat{X}_1 := \frac{\hat{\pi}_x}{eB}, \qquad \hat{P}_1 := \hat{\pi}_y \quad \hat{X}_2 := \frac{\hat{t}_y}{eB}, \qquad \hat{P}_2 := \hat{t}_x. \tag{56}$$

we may see that they satisfy the canonical commutation relation:

$$\left[\hat{X}_i, \hat{P}_j\right] = i\hbar\delta_{ij}, \quad \left[\hat{X}_i, \hat{X}_j\right] = 0, \quad \left[\hat{P}_i, \hat{P}_j\right] = 0 \tag{57}$$

Then the Hamiltonian can be written as:

$$\begin{aligned}\hat{H} &= \frac{1}{2m}\left[(eB)^2\hat{X}_1^2 + \hat{P}_1^2\right] \\ &= \frac{\hat{P}_1^2}{2m} + \frac{1}{2}\omega_c^2\hat{X}_1^2 \quad \omega_c = e\frac{B}{m}\end{aligned} \tag{58}$$

2.2.2 Landau Levels in Symmetry Gauge

We now see that the Hamiltonian is just a harmonic oscillator Hamiltonian. Thus the energy levels are quantized and given by:

$$E_n = \hbar\omega_c\left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, ... \tag{59}$$

and the eigenstates are given by:

$$\Psi_n(X_1, X_2) = \phi_n(X_1)\psi(X_2) \tag{60}$$

where $\psi(x)$ is arbitrary function and $\phi_{n(x)}$ is the n-th eigenstate of the harmonic oscillator, written explicitly as:

$$\phi_n(x) \propto H_n\left(\frac{x}{l_B}\right) \exp\left(-\frac{x^2}{2l_B^2}\right) \quad l_B = \sqrt{\frac{\hbar}{eB}} \tag{61}$$

For a general inifinite plane, the level are highy degenerate due to the fact that the wave function have infinite freedom in the X_2 direction.

2.2.3 Landau Levels in Landau Gauge

To evaluate the degeneracy of a finite system, we prefer using the Landau gauge:

$$A(x) = (0, Bx, 0) \tag{62}$$

Now the Hamiltonian is given by:

$$\hat{H} = \frac{1}{2m}\left[(\hat{p}_x)^2 + \left(\hat{p}_y - eB\hat{x}\right)^2\right] \tag{63}$$

We now take an ansatz for the wave function:

$$\Psi(x, y) = \exp\left(\frac{i}{\hbar}p_y y\right)\phi(x) \tag{64}$$

With this ansatz, the Schrodinger equation is given by:

$$\frac{1}{2m}\left[(\hat{p}_x)^2 + \left(p_y - eB\hat{x}\right)^2\right]\phi(x) = E\phi(x) \tag{65}$$

This is just the form of a harmonic oscillator, with a position shifted in x direction by $\frac{p_y}{eB}$. Thus the energy levels are still given by:

$$\begin{aligned} E_n &= \hbar\omega_c\left(n + \frac{1}{2}\right), \quad n = 0, 1, 2, ... \\ \Psi_n(x,y) &= \exp\left(\frac{i}{\hbar}p_y y\right)\phi_n(x - x_0), \quad x_0 = \frac{p_y}{eB} \end{aligned} \tag{66}$$

2.2.4 Degeneracy of Landau Levels

Now here is the set up of the system. We have a finite system with size L_x, L_y in the x and y direction. We impose:

- Periodic boundary condition in the y direction: $\Psi(x,y) = \Psi\left(x, y + L_y\right)$

This means that the momentum in the y direction is quantized:

$$p_y = \frac{2\pi\hbar}{L_y}k_y, \quad k_y \in Z \tag{67}$$

Then the LL wave function is given by:

$$\begin{aligned} &\Psi_{n,k_y}(x,y) = \exp\left(\frac{i}{\hbar}p_y y\right)\phi_n(x - x_0), \\ &x_0 = \frac{p_y}{eB}, \quad p_y = \frac{2\pi\hbar}{L_y}k_y, \quad n, k_y \in \mathbb{Z} \end{aligned} \tag{68}$$

where φ_n is the n-th eigenstate of the harmonic oscillator. If the system have finite size in the x direction, then the wave function should be localized in the system. then the maximum value of x_0 is L_x. Thus we have:

$$\frac{2\pi\hbar N_D}{eBL_y} \sim L_x \quad \Leftrightarrow \quad N_D = \frac{eBL_xL_y}{2\pi\hbar} = \frac{\Phi}{\Phi_0} \tag{69}$$

The degeneracy of each Landau level is given by the number of flux quanta through the system.

3 Lecture 7: Aharonov-Bohm Effect

3.1 Aharonov-Bohm Effect

3.2 Large Gauge Transformation

In fact, one can prove that the observable is really the phase factor not the AB phase. This lead to a rethinking of what a gauge transformation is on a non-trivial topology.

4 Lecture 8: Berry Phase

4.1 Dirac Monopole

4.1.1 Non-Global Vector Potential

Now we consider a world with a magnetic monopole. The magnetic field of a monopole is given by:

$$B(r) = \frac{e_M}{4\pi r^2}(\hat{r}), \tag{70}$$

However, due to the definition of vector potential, if we have:

$$B = \nabla \times A, \quad \nabla \cdot B = 0, \tag{71}$$

which is inconsistent with the fact that the magnetic field of a monopole has a non-zero divergence. To solve this problem, we can assume that the vector potential **is not globally defined**.

4.1.2 Dirac Quantiztion

YL: [The argument during the class is not understandable. I'll not think like this. Don't even try to understand this, it only makes things more confusing.]

Here I take the argument in the note of David Tong.

4.2 Berry Phase

4.2.1 Global Phase Transformation Invariant

We first want to have a well defined phase. Notice that quantum states are **rays** in the Hilbert space, and doing a global phase transformation does not change the physical state. For a series of states:

$$|\psi_1\rangle, |\psi_2\rangle, ..., |\psi_N\rangle \tag{72}$$

we can always perform a global phase transformation:

$$|\psi_{i,\Lambda}\rangle = \exp(i\Lambda_i)\ |\psi_i\rangle, \quad \Lambda \in R \tag{73}$$

and the states are unchanged. Thus the phase of a quantum state is not well defined. Then we have a look if the phase difference

Bibliography